

What is HTML?

- ▶ **HTML** stands for **HyperText Markup Language**.
- ▶ **HTML** is the **standard markup language** used to create the **structure** of web pages.
- ▶ It is not a programming language – it **describes the content and layout** of a page using elements and tags.
- ▶ To structure content on the web.
- ▶ To embed elements such as headings, paragraphs, images, videos, and links.
- ▶ To work together with CSS (for design) and JavaScript (for interaction).

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My First Page</title>
</head>
<body>
  <h1>Hello, World!</h1>
  <p>This is my first HTML page.</p>
</body>
</html>
```

HTML

- ▶ `<!DOCTYPE html>` → Tells the browser this is an HTML5 document.
- ▶ `<html>` → Root element of the page.
- ▶ `<head>` → Contains metadata (title, character set, CSS links, etc.).
- ▶ `<title>` → Text shown in the browser tab.
- ▶ `<body>` → Visible content (headings, text, images, links, etc.).

Key HTML Attributes

- ▶ `id` → Unique identifier for an element.
- ▶ `class` → Groups multiple elements for styling.
- ▶ `src` → Specifies source (for images, videos, etc.).
- ▶ `href` → Link URL for anchor tags.
- ▶ `alt` → Alternative text for images.
- ▶ `style` → Inline CSS styling.
- ▶ Attributes give extra information to elements.
- ▶ Written inside opening tags.

Advantages of HTML

- ▶ Easy to learn.
- ▶ Supported by all browsers.
- ▶ Integrates with CSS & JavaScript.
- ▶ Used to build any kind of website.
- ▶ HTML is completely free; no license or cost is required.
- ▶ Supported by all browsers, making it universally accessible.
- ▶ HTML works on any operating system (Windows, Linux, macOS) and any browser.
- ▶ Web pages created in HTML can be viewed across multiple devices without modification.
- ▶ HTML files are small in size, so pages load quickly.
- ▶ Minimal resource usage makes it ideal for both low- and high-speed internet connections.

Limitations of HTML

- ▶ HTML creates static web pages.
- ▶ It cannot perform dynamic tasks like interacting with databases or responding to user input without other technologies (e.g., JavaScript, PHP).
- ▶ HTML alone cannot provide advanced styling or layout control.
- ▶ Requires CSS for better design and presentation.
- ▶ HTML cannot handle programming logic (loops, conditions, calculations).
- ▶ For dynamic behavior, scripting languages like JavaScript are needed.
- ▶ HTML cannot protect data or implement authentication.
- ▶ HTML alone cannot store or retrieve data.

Characteristics of HTML

- ▶ Simple & Easy – Easy to learn and use.
- ▶ Platform Independent – Works on all browsers (Chrome, Edge, Firefox, etc.).
- ▶ Extensible – Supports multimedia (images, audio, video).
- ▶ Linking – Allows linking to other pages using hyperlinks.
- ▶ Structural – Provides structure (skeleton) to a website.

What is HTML?

Version	Features
HTML 1.0	Basic formatting
HTML 2.0	Basic structure and forms
HTML 3.2	Tables and scripting
HTML 4.01	Better structure, support for CSS
HTML5	Modern version – supports video , audio , canvas , local storage , responsive design etc.

Importance of HTML

- ▶ HTML is the foundation of all websites.
- ▶ Without HTML, a web browser cannot understand or render any content.
- ▶ Every web developer must know HTML as the starting point for web development.
- ▶ **Use Cases**
 - ▶ Designing webpages
 - ▶ Creating email templates
 - ▶ Structuring documents in web apps
 - ▶ Building layouts for CMS systems (like WordPress, Magento)

HTML Fundamentals

- ▶ HTML is not a programming language — it's a markup language.
- ▶ `<!DOCTYPE html>` = HTML5 declaration
- ▶ `<html>` = Root tag
- ▶ `<head>` = Metadata
- ▶ `<body>` = Content visible on the page
- ▶ `<div>` = Block-level container (used to group elements)
- ▶ `<span>` = Inline-level container

HTML Fundamentals

Tag	Purpose	Example	
<code><html></code>	Root of HTML document	<code><html> ... </html></code>	
<code><head></code>	Contains metadata, title, links, etc.	<code><head> ... </head></code>	
<code><title></code>	Page title (shown in browser tab)	<code><title>My Page</title></code>	
<code><body></code>	Visible content	<code><body> ... </body></code>	
<code><h1></code> to <code><h6></code>	Headings (h1 = biggest, h6 = smallest)	<code><h1>Hello</h1></code>	
<code><p></code>	Paragraph	<code><p>This is a paragraph.</p></code>	
<code><a></code>	Link	<code>Google</code>	
<code></code>	Image	<code></code>	



HTML Fundamentals



<code>
</code>	Line break	<code>Line 1
Line 2</code>
<code><hr></code>	Horizontal line	<code><hr></code>
<code></code>	Unordered list	<code>Item</code>
<code></code>	Ordered list	<code>First</code>
<code></code>	List item	<code>Apple</code>
<code><div></code>	Block container	<code><div>Content here</div></code>
<code></code>	Inline container	<code>Text</code>



The List Tags

- ▶ What are List Tags in HTML?
- ▶ In HTML, **lists** are used to display multiple items in a structured way. Lists help organize content into **points** so it's easier for readers to understand.
Each list type has a **parent tag** and **child list item tags**.
- ▶ Types of Lists
- ▶ An ordered list displays items in a specific sequence.
- ▶ Items are numbered automatically by the browser.
- ▶ Used when the order of items matters.
- ▶ Each item in the list is written inside `<li>` (List Item) tags.

The List Tags

► Example

```
<ol>  
  <li>Step 1: Open the browser</li>  
  <li>Step 2: Type the website address</li>  
  <li>Step 3: Press Enter</li>  
</ol>
```

- type – Changes numbering style
- 1 → Numbers (1, 2, 3)
- A → Capital letters (A, B, C)
- a → Small letters (a, b, c)
- I → Capital Roman (I, II, III)
- i → Small Roman (i, ii, iii)

The List Tags

- ▶ start – Sets the starting number.
- ▶ Example: `<ol start="5">` starts from 5.

Unordered List

- ▶ B) Unordered List ()
- ▶ An unordered list displays items with bullets instead of numbers.
- ▶ Used when the order of items does not matter.
- ▶ Each item is also written inside tags.

```
<ul>  
  <li>Milk</li>  
  <li>Bread</li>  
  <li>Eggs</li>  
</ul>
```


Unordered List

- ▶ Styling Bullets:
- ▶ Using the list-style-type property in CSS, you can change bullet styles:
- ▶ disc (● – default)
- ▶ circle (○)
- ▶ square (■)
- ▶ none (no bullets)

Definition List (<dl>)

- ▶ Definition List (<dl>)
- ▶ A definition list is used for terms and their meanings.
- ▶ Contains:
- ▶ <dt> → Definition Term (word or phrase being defined)
- ▶ <dd> → Definition Description (explanation of the term)

```
<dl>  
  <dt>HTML</dt>  
  <dd>- HyperText Markup Language</dd>  
  
  <dt>CSS</dt>  
  <dd>- Cascading Style Sheets</dd>  
</dl>
```

Definition List (<dl>)

- ▶ **Advantages of Using Lists**
- ▶ Makes content **more organized** and **easy to read**.
- ▶ Improves **visual presentation**.
- ▶ Helps in **step-by-step instructions**.
- ▶ Search engines can better understand **structured content**.

What is an HTML Table?

- ▶ An HTML table is used to display data in a row and column format. Tables are useful for organizing data such as schedules, reports, or lists.
- ▶ **Why Use HTML Tables?**
- ▶ Organize structured data (like rows/columns in Excel).
- ▶ Display tabular information (reports, lists, price charts).
- ▶ Create layouts (less common now due to CSS Flex/Grid, but used historically).

Frames in HTML

- ▶ A frame is a part of a web page that can display another HTML page independently.
- ▶ Frames allow you to divide a browser window into multiple sections, each showing a different HTML document.
- ▶ `<iframe>` → Inline frame that embeds another page.
- ▶ `src` → URL of the page to display.
- ▶ `width & height` → Size of the iframe.
- ▶ `title` → For accessibility.
- ▶ **Advantages of Frames / Iframes**
- ▶ Allows multiple HTML pages in a single window.
- ▶ Can keep a menu or header constant while content changes.
- ▶ Useful for embedding videos, maps, ads.

Frames in HTML

- ▶ Frames allow a web browser window to display multiple HTML documents simultaneously.
- ▶ Each section of the window is called a frame.
- ▶ Frames are mainly used to keep navigation menus, headers, or sidebars fixed while the content in another frame changes.

Basic Example of Frames

```
<!DOCTYPE html>
<html>
<head>
    <title>Frames Example</title>
</head>
<frameset cols="50%,50%"> <!-- Divide window into 2
    <frame src="page1.html" name="leftFrame">
    <frame src="page2.html" name="rightFrame">
</frameset>
</html>
```

Frames in HTML

```
<!DOCTYPE html>
<html>
<head>
  <title>Iframe Example</title>
</head>
<body>
  <h1>My Web Page</h1>
  <iframe src="page1.html" width="600" height="400" title="My Frame"></iframe>
</body>
</html>
```


What is an HTML Table?

Tag	Description
<table>	Starts the table.
<tr>	Table row: groups a set of cells into a row.
<th>	Table header cell: bold and centered by default.
<td>	Table data cell: holds the actual content.
<caption>	Adds a title above the table.
<thead>	Groups the heading part (rows of <th>).
<tbody>	Groups the main content rows.
<tfoot>	Groups footer content (e.g., totals).

Example

```
<table>
  <tr><th>Name</th><th>Role</th></tr>
  <tbody>
    <tr><td>Amit</td><td>Designer</td></tr>
    <tr><td>Rita</td><td>Developer</td></tr>
  </tbody>
</table>
```

What is a Form?

- ▶ An HTML form is used to collect user input on a web page.
- ▶ Users can enter text, select options, click buttons, etc., and submit the data to a server.
- ▶ Forms are essential for login pages, registration pages, surveys, search boxes, etc.
- ▶ Use cases
 - ▶ Registration/login forms
 - ▶ Contact forms
 - ▶ Surveys and feedback
 - ▶ Online orders and shopping carts

What is a Form?

Input Type	Description
text	Single line text
password	Masked text for passwords
email	Validates email format
number	Accepts numeric input
tel	Telephone number
url	Validates URL format
date	Date picker
radio	Single choice among options
file	Upload files
range	Slider for number selection
checkbox	Multiple selections

What is a Form?

- ▶ required → Field must be filled.
- ▶ pattern → Regex pattern to validate input.
- ▶ min / max → Minimum/maximum value for numbers or dates.
- ▶ maxlength → Maximum number of characters.
- ▶ type → Built-in validation (email, url, number).
- ▶ Forms are submitted using `<button type="submit">` or `<input type="submit">`.
- ▶ After submission, the browser navigates to the URL in the action attribute.
- ▶ JavaScript can be used to validate or intercept the submission.

What is a Form?

```
<form action="submit.php" method="post" enctype="multipart/form-data">
  <!-- Text Input -->
  <label for="name">Full Name:</label>
  <input type="text" id="name" name="fullname" placeholder="Enter your full name"
  required>
  <!-- Password Input -->
  <label for="password">Password:</label>
  <input type="password" id="password" name="password" placeholder="Enter password"
  required>
  <!-- Email Input -->
  <label for="email">Email:</label>
  <input type="email" id="email" name="email" placeholder="Enter email" required>
  <!-- Number Input -->
  <label for="age">Age:</label>
  <input type="number" id="age" name="age" min="1" max="100" placeholder="Enter your
  age" required>
```

What is a Form?

```
<!-- Telephone Input -->
<label for="phone">Phone Number:</label>
<input type="tel" id="phone" name="phone" placeholder="Enter phone number"
"[0-9]{10}" required>
<!-- URL Input -->
<label for="website">Website:</label>
<input type="url" id="website" name="website" placeholder="https://example.com" required>
<!-- Date Input -->
<label for="dob">Date of Birth:</label>
<input type="date" id="dob" name="dob" required>
<!-- Time Input -->
<label for="appt">Preferred Time:</label>
<input type="time" id="appt" name="appt">
<!-- Color Input -->
<label for="color">Favorite Color:</label>
<input type="color" id="color" name="color">
<!-- Range Input -->
<label for="rating">Rating (1-10):</label>
<input type="range" id="rating" name="rating" min="1" max="10">
<!-- Textarea -->
```

Features in This Form

- ▶ All major input types: text, password, email, number, tel, url, date, time, color, range.
- ▶ Textarea, select (dropdown).
- ▶ Radio buttons and checkboxes inside `<fieldset>` with `<legend>`.
- ▶ File upload and hidden input.
- ▶ Submit and reset buttons.
- ▶ Basic styling for readability.
- ▶ HTML5 validation: required, pattern, min, max.